

Word Problems with Metric Measurements

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Word Problems with Metric Measurements

On the GED math test, you will be expected to know the basic conversions that you have learned in this lesson. If any other measurement fact is needed to solve a problem, the information will be provided with the test item. You can create a unit multiplier from any fact that you are given. A **unit multiplier** is a fraction, written with two different measurement units, that has a value of one.

Example

Hide

Justin needs to know how many inches are equal to 1 meter. He is given the following information: 1 in = 2.54 cm.

Step 1

Justin needs to write a chain of unit multipliers that connects meters and inches. In addition to $1 \text{ in} = 2.54 \text{ cm}$, he needs another fact that connects centimeters and meters. He uses the fact that $100 \text{ cm} = 1 \text{ m}$.

$$\frac{1 \text{ in}}{2.54 \text{ cm}} \times \frac{100 \text{ cm}}{1 \text{ m}}$$

Step 2

Multiply. Cancel like measurement units.

$$\frac{1 \text{ in}}{2.54 \text{ cm}} \times \frac{100 \cancel{\text{cm}}}{1 \text{ m}} = \frac{100 \text{ in}}{2.54 \text{ m}}$$

Step 3

Divide to create a unit rate.

$$100 \div 2.54 \approx 39.37 \text{ inches per meter}$$

There are about **39.37 inches** in 1 meter.

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^ High

= Medium

Low

Category

Lesson 2: The Metric System

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